



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Design Thinking Course for Businesses

TGS Ref No: TGS-2026064719

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 15.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
14.0	1 January 2024	Previous release under course code TGS-2020503676 — 1-day lesson plan for the Design Thinking Course for Businesses.	Dr Alfred Ang
15.0	10 August 2026	Re-issued under WSQ course code TGS-2026064719. Content carried over from the v14 master trainer deck and restructured to the current Tertiary Infotech WSQ house standard. Added seven structured hands-on activities built on the browser-based ed-tools (Design Thinking Toolkit, RACI, Scrum, Digital Transformation and BCM), and aligned the schedule, slide references and assessment instruments (WA SAQ + PP).	Dr Alfred Ang

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Course Information

Course Title	Design Thinking Course for Businesses
WSQ Course Reference	TGS-2026064719
TSC Title / Code	Design Thinking Practices · DSN-ACE-3014-1.1
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	1 day · 8 course hours (375 minutes instruction + 105 minutes assessment; the 1-hour lunch break is excluded)
Daily Timing	9:30 am – 6:30 pm, including a 1-hour lunch break; tea breaks are taken within training time
Mode	Instructor-led, facilitated hands-on design-thinking activities using browser-based ed-tools
Assessment	Written Assessment (WA — SAQ, 30 minutes) and Practical Performance (PP, 70 minutes), both open book
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Understand key concepts of design thinking to communicate the design outcomes
- LO2: Apply design thinking to generate new ideas for the organization
- LO3: Uncover opportunities of design thinking
- LO4: Implement plans to embed various stages of design thinking across the organization
- LO5: Execute the design concept through prototypes
- LO6: Utilize metrics to measure the outcomes of design ideas and solutions

Skills Framework Alignment

This course is aligned to the Skills Framework for Design, TSC "Design Thinking Practices" (DSN-ACE-3014-1.1).

Abilities

- A1: Apply design thinking methodologies to define design problems and generate new ideas for the organization
- A2: Uncover opportunities for applying design thinking across the organization
- A3: Utilize metrics to benchmark and measure outcomes of design ideas and solutions
- A4: Implement plans to embed design thinking across the organization
- A5: Facilitate the development of design concepts through prototypes and visual tools
- A6: Communicate design outcomes and their value to stakeholders

Knowledge

- K1: Key concepts and principles of design thinking
- K2: Importance and business value of design thinking
- K3: Traits and mindset of a design thinker
- K4: Action phases of design thinking and the activities within each phase
- K5: Use cases and applications of design thinking across industries
- K6: Approaches to applying design thinking within an organization
- K7: Methodologies, visual tools and metrics used at each action phase

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 30 minutes, open book.
- Practical Performance (PP) — design-thinking tasks using the course ed-tools, 70 minutes, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- The final assessment is conducted at the end of Day 1, after the course recap and TRAQOM survey.
- Learners must be assessed as 'Competent' in both instruments to be awarded the Statement of Attainment.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Ed-Tools Used in the Activities

All activity tools are browser-based and require no installation:

Ed-Tool	Link	Used for
Design Thinking Toolkit	https://alfredang.github.io/designthinking/	Empathy Map, Persona, Journey Map, POV, HMW, Brainstorm, Prototype and Test canvases — the primary tool for this course.
RACI Matrix	https://alfredang.github.io/raci/	Assign Responsible / Accountable / Consulted / Informed roles so a design idea has a clear owner when it moves to delivery.
Scrum Planner	https://alfredang.github.io/scrum/	Turn validated design concepts into a prioritised product backlog and sprint plan.
Digital Transformation Canvas	https://alfredang.github.io/digitaltransformation/	Position a design initiative within the organisation's wider transformation roadmap.
Business Continuity Management	https://alfredang.github.io/bcm/	Stress-test a design solution for operational risk and continuity before rollout.

Course Schedule

Day 1 — Design Thinking Concepts, Applications, Action Phases, Visual Tools & Assessment

Time	Duration	Topic / Activity
9:30–9:50	20 min	Welcome, trainer and learner introductions, ground rules, course outline, learning outcomes and mandatory SSG digital attendance (AM). Download of course material from the LMS/TMS portal and introduction to the browser-based ed-tools used in the activities
9:50–10:35	45 min	Topic 1 — Key Concepts & Principles of Design Thinking: what design thinking is, why it matters, ROI and business value, analytical vs creative thinking, the design thinker's mindset, growth mindset, and the desirability/viability/feasibility lenses (concepts + facilitated discussion)
10:35–10:50	15 min	Tea break (within training time)
10:50–11:30	40 min	Hands-on: Activity 1: Design a Vase — Reframing the Brief; Activity 2: Design Thinking Readiness — Mindset & Culture Self-Assessment
11:30–12:25	55 min	Topic 2 — Applications of Design Thinking: corporate, service and public-sector use cases (Airbnb, IBM, Bank of America, Uber Eats, healthcare, Clean Team, GGRC, The Good Kitchen) and how to uncover opportunities. Hands-on: Activity 3: Uncover Design Thinking Opportunities in Your Business
12:25–1:25	60 min	Lunch break
1:25–2:05	40 min	Topic 3 — Action Phases of Design Thinking: the Stanford d.school five-stage model — Empathize, Define, Ideate, Prototype and Test — including objectives, techniques and the iterative nature of the loop. Mandatory SSG digital attendance (PM)
2:05–2:45	40 min	Hands-on: Activity 4: The

		Wallet Project — A Full Design Thinking Cycle
2:45–3:00	15 min	Tea break (within training time)
3:00–3:35	35 min	Topic 4 — Methodologies and Visual Tools: empathy maps, personas and journey maps; POV and How Might We; brainstorming and idea-selection methods; low- and high-fidelity prototyping; user testing; and metrics for measuring design outcomes
3:35–4:00	25 min	Hands-on: Activity 5: Empathy Map & Persona — The Gift-Giving Experience
4:00–4:35	35 min	Hands-on: Activity 6: POV, How Might We & Brainstorming; Activity 7: Prototype, Test & Measure the Outcome
4:35–4:45	10 min	Course recap, summary and Q&A, course feedback and the mandatory TRAQOM survey
4:45–4:50	5 min	Briefing for Assessment and Assessment digital attendance
4:50–5:20	30 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 30 minutes, open book
5:20–6:30	70 min	Practical Performance (PP) — design-thinking tasks using the course ed-tools, 70 minutes, open book

Total course hours: 480 minutes (8 hours) — 375 minutes of instruction (including tea breaks taken within training time) plus 105 minutes of assessment. The 1-hour lunch break is excluded.

Activity Reference (aligned to topics and slides)

Topic / Skills mapped	Slide positions	Activities	Learning outcome
Topic 01: Key Concepts & Principles of Design Thinking (A6, K1, K2, K3, A1, K6)	Slides 17–33	Activity 1, Activity 2	LO1; LO1, LO4
Topic 02: Applications of Design Thinking (A2, K5)	Slides 34–42	Activity 3	LO3
Topic 03: Action Phases of Design Thinking (A4, K4, A1)	Slides 43–58	Activity 4	LO2, LO4, LO5
Topic 04: Methodologies and Visual Tools for Design Thinking	Slides 59–78	Activity 5, Activity 6, Activity 7	LO2; LO2, LO3; LO5, LO6

(A5, K7, A3)			
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Detailed Activity Plan

Activity	Learners will...	Ed-tool	Trainer assesses
Activity 1 Design a Vase — Reframing the Brief	A two-round warm-up that shows the difference between designing an object and designing an experience. In round 1 you design a vase. In round 2 you re-design against a human-centred brief and compare how much wider the solution space becomes.	https://alfredang.github.io/designthinking/	You have two visibly different concepts on your board, and you can state in one sentence why the second brief produced a wider range of ideas than the first.
Activity 2 Design Thinking Readiness — Mindset & Culture Self-Assessment	Benchmark your own team against the traits of a design thinker and the three lenses of organisational design thinking — desirability, viability and feasibility — then identify the single biggest barrier to embedding design thinking where you work.	https://alfredang.github.io/digitaltransformation/	Your scorecard is complete across all three lenses, and you have named one barrier and one specific, time-bound action to address it.
Activity 3 Uncover Design Thinking Opportunities in Your Business	Study how Airbnb, IBM, Bank of America, Uber Eats and the public-sector cases applied design thinking, extract the transferable pattern from each, then map three concrete opportunities in your own business and rank them by impact and effort.	https://alfredang.github.io/designthinking/	You can name three design-thinking opportunities in your own organisation, each with an identified user group and observable symptom, and justify which one you would tackle first.
Activity 4 The Wallet Project — A Full Design Thinking Cycle	The classic Stanford d.school immersive exercise. Working in pairs, you take a partner through all five action phases — Empathize, Define, Ideate, Prototype and Test — to redesign their wallet, or more precisely the experience of carrying what matters to them. It is deliberately fast so you feel the rhythm of the whole cycle.	https://alfredang.github.io/designthinking/	You have completed all five phases, your prototype changed as a direct result of your partner's test feedback, and you can state the POV insight in your partner's own words.
Activity 5 Empathy Map & Persona — The Gift-	Redesign the gift-giving experience for a partner. You start by	https://alfredang.github.io/designthinking/	All four quadrants are populated with your partner's own words,

Giving Experience	building empathy: interview them about the last gift they gave, then plot what they said, thought, did and felt into a four-quadrant empathy map and distil it into a persona.		you have identified at least one contradiction between quadrants, and your partner confirms the persona sounds like them.
Activity 6 POV, How Might We & Brainstorming	Take the empathy map from the previous lab and convert it into a sharp Point of View problem statement, reframe it as a set of 'How Might We' questions, then run a timed brainstorm and converge on the best ideas using dot voting.	https://alfredang.github.io/designthinking/	Your POV has a verb-based need and a genuine insight, you generated 20 or more ideas, and you have three shortlisted ideas each with a stated reason for selection.
Activity 7 Prototype, Test & Measure the Outcome	Turn your shortlisted idea into a low-fidelity prototype, test it with a real user, then define how you would measure whether it succeeded — moving from 'we like this idea' to 'here is the evidence it works'. Finish by handing the concept over for delivery with a RACI matrix and a sprint plan.	https://alfredang.github.io/designthinking/	Your prototype was changed by user feedback, you have named the assumption you tested and the metrics that would prove success, the concept has an owner in the RACI matrix and a first sprint goal, and you have pitched the outcome and its value to your stakeholder leading with the insight rather than the solution.

Trainer Notes

- The deck is deliberately visual and carries no step-by-step instruction slides — walk learners through the steps using the Learner Guide and the labs/ files while they work.
- Activity 1 (Design a Vase) is the key hook of the morning: run round 1 strictly to 2 minutes so the narrow-brief effect is felt before you reveal the reframed brief.
- Activity 4 (the Wallet Project) is the only full five-phase cycle. Keep each phase to time and enforce the swap so both partners design.
- Ed-tools are browser-based; verify learners can reach alfredang.github.io before the first activity and have a paper fallback ready.
- Collect evidence for the Practical Performance assessment as learners work — their empathy map, POV statement and prototype form the basis of the PP tasks.
- Digital attendance must be taken three times: AM, PM and Assessment. Remind learners of the 75% attendance rule for funding eligibility.